Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Montorio, Nico Adrielle N.

BSIT-4A

PART 1 - COMPANY PROFILE

1.1 Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a newly established software development and IT consulting company that provides customized software solutions, web and application development, database management, and technical support services to small and medium-sized businesses. The company focuses on developing reliable, secure, and scalable digital solutions that help organizations improve their business operations and productivity.

1.3 Company Vision

To become a trusted technology partner for businesses by delivering innovative, secure, and reliable software solutions that contribute to digital transformation and long-term growth.

1.4 Company Mission

ABC Startup Solutions is committed to providing high-quality software development and IT services through innovative technologies, skilled professionals, and customer-focused solutions. The company aims to maintain secure and efficient IT operations while continuously improving its services to meet the changing needs of its clients.

1.5 Office Location

Location: 3rd Floor, ABC Business Center, Purok 3, Santa Rosa, Laguna, Philippines

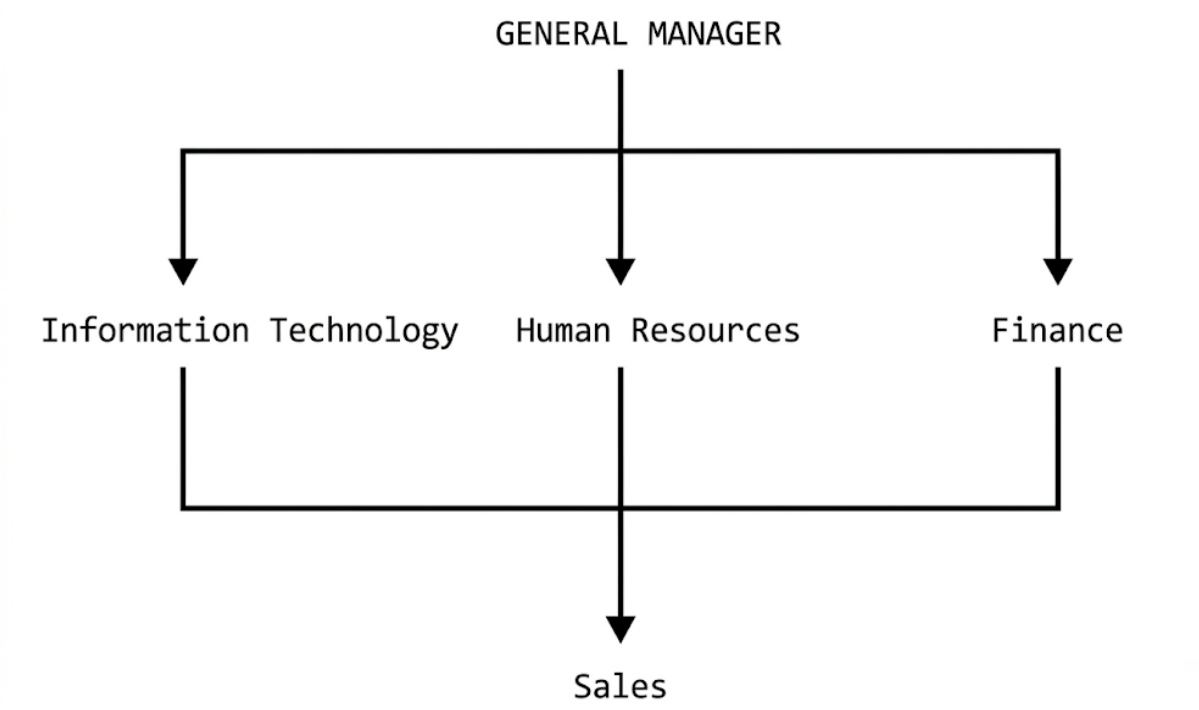
The company operates from a single office floor designed to accommodate its 20 employees, development activities, administrative operations, client coordination, and IT infrastructure.

1.6 Organizational Structure

ABC Startup Solutions uses a simple organizational structure appropriate for a newly established software development company. The General Manager oversees the overall business operations and coordinates with the four primary departments: Information Technology, Human Resources, Finance, and Sales.

The Information Technology Department is responsible for software development, technical support, system administration, network management, and IT infrastructure. The Human Resources Department manages recruitment, employee records, training, and company policies. The Finance Department handles budgeting, accounting, payroll, and financial records. The Sales Department manages client relationships, marketing, sales activities, and business development.

Organizational Structure



For the company's day-to-day operations, each department works collaboratively while maintaining responsibility for its specific business functions.

1.7 Employee Distribution

ABC Startup Solutions has a total of 20 employees distributed among four departments.

Department	Number of Employees	Percentage
Information Technology	5	25%
Human Resources	4	20%
Finance	5	25%
Sales	6	30%
Total	**20**	**100%**

Department Functions

Department	Primary Function
Information Technology	Software development, IT support, system administration, network management, and infrastructure maintenance
Human Resources	Recruitment, employee records, training, benefits, and personnel management
Finance	Accounting, payroll, budgeting, financial reporting, and financial record management
Sales	Client acquisition, sales management, marketing, customer relationships, and business development

1.8 Company IT Requirements

Because ABC Startup Solutions is a software development company, reliable IT infrastructure is essential to its daily operations. Employees require computers, network connectivity, productivity software, secure access to company resources, and reliable data storage. The Information Technology Department also requires additional resources for software development, server administration, network management, testing, and technical support.

The company's infrastructure must therefore provide **reliable connectivity, centralized resources, data protection, security, scalability, and backup capabilities**. These requirements will serve as the foundation for the hardware, software, networking, and security recommendations presented in the succeeding sections of this infrastructure plan.

PART 2 - ENTERPRISE HARDWARE INVENTORY

2.1 Hardware Inventory Overview

ABC Startup Solutions requires reliable hardware to support its 20 employees and daily business operations. The hardware inventory is designed according to the needs of each department while considering performance, security, availability, and future expansion.

The company will use a combination of desktop computers and laptops. Desktop computers will be assigned primarily to employees who work from fixed office locations, while laptops will provide mobility for IT personnel, management, and employees who may need to work in different areas of the office or attend client meetings.

A dedicated server, NAS storage, UPS units, network equipment, printers, and backup drives will provide the infrastructure required for centralized services, data storage, business continuity, and network operations.

2.2 Hardware Inventory Table

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	14	IT, HR, Finance, Sales	Primary workstation for employees performing daily office and development tasks
HW-002	Business Laptop	6	IT, Management, Sales	Portable workstation for IT staff, management activities, client meetings, and mobile work
HW-003	24-inch LED Monitor	20	All Departments	Provides a dedicated display for each employee workstation
HW-004	Rackmount Server	1	IT	Hosts centralized services, file sharing, internal applications, user management, and other server-based services
HW-005	Business Router	1	IT	Manages network routing, DHCP, and communication between internal networks and the internet

HW-006	Managed Gigabit Switch	1	IT	Connects wired computers, servers, printers, access points, and other network devices
HW-007	Network Firewall	1	IT	Protects the internal network from unauthorized access and external threats
HW-008	Wireless Access Point	2	IT	Provides secure wireless network connectivity throughout the office
HW-009	Network Printer	2	HR, Finance, Sales	Provides shared printing for documents, reports, invoices, and administrative records
HW-010	UPS	2	IT	Provides temporary power and protection for the server and critical network equipment
HW-011	NAS Storage	1	IT	Provides centralized file storage and backup storage for company data
HW-012	External Backup Drive	2	IT	Provides offline backup copies of important company data
HW-013	Network Patch Panel	1	IT	Organizes and manages network cable connections inside the network rack
HW-014	CAT6 Ethernet Cable	30	All Departments	Provides reliable wired network connections for computers and network equipment
HW-015	RJ45 Connectors	50	IT	Used for terminating and repairing Ethernet network cables
HW-016	Network Rack	1	IT	Houses and organizes the server, switch, patch panel, UPS, and other network equipment

2.3 Hardware Quantity Justification

Desktop Computers – 14 Units

Fourteen desktop computers will be provided for employees whose work primarily takes place at their assigned office workstations. These computers will be distributed among IT, HR, Finance, and Sales based on departmental requirements.

The IT department may use higher-performance workstations for software development, testing, and system administration, while HR, Finance, and Sales employees can use business-class systems suitable for office applications and web-based services.

Business Laptops – 6 Units

Six laptops will provide mobility for employees who need to work away from their permanent desks. Laptops will primarily be assigned to selected IT personnel, management, and Sales employees who may attend client meetings or perform presentations.

The combination of desktops and laptops provides the company with both performance and flexibility without requiring every employee to use a laptop.

Monitors – 20 Units

Twenty 24-inch monitors are recommended so that every employee has a dedicated display. Additional screen space is particularly useful for IT personnel who may need to work with source code, documentation, terminals, databases, and development tools simultaneously.

Server – 1 Unit

One dedicated server is sufficient for the company's initial 20-employee operation. The server can provide centralized services such as file sharing, internal applications, user management, and other infrastructure services.

The server should have sufficient processing power, memory, storage capacity, and redundancy to support the company's initial requirements while allowing future expansion.

Router – 1 Unit

One business-class router will manage communication between the company's internal network and the internet. It will also support functions such as DHCP, routing, and network management.

Managed Switch – 1 Unit

A managed switch will connect the company's wired devices to the network. A switch with sufficient ports should be selected to accommodate the 20 employee workstations, server, printers, access points, NAS, and future devices.

A managed switch is preferred because it supports features such as VLANs, monitoring, and traffic management.

Firewall – 1 Unit

A dedicated firewall will provide an additional layer of network security. It will help control incoming and outgoing traffic, restrict unauthorized access, and provide security monitoring.

The firewall is particularly important because the company will handle business information, employee records, financial information, source code, and client data.

Wireless Access Points – 2 Units

Two wireless access points are recommended to provide reliable Wi-Fi coverage throughout the single-floor office. Using two access points can reduce dead zones and distribute wireless clients more effectively.

Separate wireless networks can also be configured for employees and guests.

Network Printers – 2 Units

Two network printers will be shared among departments rather than assigning a printer to every employee. This reduces equipment and maintenance costs while still providing convenient access to printing.

One printer can primarily support HR and Finance, while the other can support Sales and general office requirements.

UPS – 2 Units

Two UPS units will protect critical infrastructure from power interruptions and electrical fluctuations.

One UPS can support the server while another protects important network equipment such as the firewall, router, and switch. This helps prevent sudden shutdowns and reduces the risk of data corruption or hardware damage.

NAS Storage – 1 Unit

A NAS device will provide centralized network storage and backup capabilities. It can be used to store shared files and backup copies of important company information.

The NAS should support RAID to improve storage availability in case of a drive failure.

External Backup Drives – 2 Units

Two external backup drives will be maintained as offline backup media. Keeping backup drives disconnected when they are not being used provides additional protection against ransomware and other network-based threats.

The drives should be rotated regularly and stored securely.

Network Rack – 1 Unit

A network rack will provide an organized and secure location for the company's server and networking equipment. Centralizing the infrastructure makes maintenance and troubleshooting easier.

CAT6 Cables and RJ45 Connectors

CAT6 Ethernet cables will be used for the company's wired network because they provide reliable Gigabit Ethernet connectivity and sufficient performance for a small enterprise environment.

Additional cables and RJ45 connectors will be maintained as spare materials for future installations, repairs, and network expansion.

2.4 Recommended Hardware Specifications

Hardware	Recommended Minimum Specification
Desktop Computer	Intel Core i5 / AMD Ryzen 5, 16 GB RAM, 512 GB SSD, Gigabit Ethernet
Business Laptop	Intel Core i5 / AMD Ryzen 5, 16 GB RAM, 512 GB SSD, Wi-Fi 6
Monitor	24-inch Full HD LED monitor
Server	8-core CPU, 32–64 GB ECC RAM, RAID-capable storage, dual Gigabit Ethernet
Router	Business-class router with VLAN, DHCP, VPN, and firewall support
Managed Switch	24-port Gigabit managed switch with VLAN support
Firewall	Business firewall with VPN, intrusion prevention, traffic filtering, and logging
Wireless AP	Wi-Fi 6 business access point
NAS	4-bay NAS with RAID support and network backup capabilities
UPS	1000–1500 VA depending on connected equipment
Network Printer	Network-enabled laser printer with automatic duplex printing
External Backup Drive	2–4 TB external storage drive
Network Rack	12U–18U rack suitable for the company's initial infrastructure

2.5 Hardware Procurement Considerations

The company should prioritize **business-grade hardware** rather than consumer-grade equipment because the infrastructure will support daily business operations and company data.

Hardware should also have manufacturer warranties, available replacement parts, and adequate technical support. Equipment should be selected with future expansion in mind so that additional employees and devices can be accommodated without requiring a complete infrastructure replacement.

The recommended inventory provides ABC Startup Solutions with the essential hardware required to establish its first office network while maintaining reasonable capacity for future growth.

PART 3 – ENTERPRISE SOFTWARE INVENTORY

3.1 Software Inventory Overview

ABC Startup Solutions requires a combination of operating systems, productivity applications, development tools, security software, and system administration utilities to support its 20 employees.

The selected software is based on the company's nature as a software development and IT consulting business. Software must support daily office operations, software development, system administration, cybersecurity, file management, and remote technical support.

Where possible, the company should use properly licensed software and open-source solutions to balance functionality, security, and operating costs.

Software	Version	License	Purpose
Windows 11 Pro	Current supported version	Commercial	Primary operating system for employee desktop computers and laptops
Ubuntu Server	Current LTS version	Open Source	Server operating system for hosting internal services and application
Microsoft 365 / Microsoft Office	Current version	Subscription / Commercial	Word processing, spreadsheets, presentations, email, and business productivity
Visual Studio Code	Current version	Free / Proprietary License	Source-code editing and software development
Git	Current stable version	Open Source	Version control and source-code management
GitHub Desktop	Current version	Free	Graphical interface for managing Git repositories
VirtualBox	Current version	Open Source	Virtual machines for testing operating systems and software
Google Chrome	Current stable version	Free / Proprietary	Web browsing and access to web-based business applications

Microsoft Defender	Included with Windows 11 Pro	Included / Commercial	Malware protection, threat detection, and endpoint security
AnyDesk	Current version	Included / Commercial	Remote desktop access and technical support
7-Zip	Current version	Open Source	File compression, extraction, and archive management

3.3 Software Purpose and Justification

Windows 11 Pro

Windows 11 Pro will be the primary operating system for employee workstations. The Pro edition is appropriate for a business environment because it includes features designed for organizational management and security.

It will be installed on the company's desktop computers and business laptops. The IT Department can also use its management and security features to maintain company devices.

Ubuntu Server

Ubuntu Server will be used as the primary server operating system for the company's server infrastructure. Its stability, security features, long-term support releases, and lack of licensing fees make it suitable for a startup with a limited initial IT budget.

The IT Department can use Ubuntu Server to host internal applications, file services, development services, databases, or other infrastructure services as required.

Microsoft 365 / Microsoft Office

Microsoft 365 provides employees with essential productivity applications such as Word, Excel, and PowerPoint. These applications are particularly important for HR documentation, financial spreadsheets, sales presentations, reports, and general business correspondence.

Microsoft 365 can also provide cloud-based collaboration and productivity services depending on the subscription selected by the company.

Visual Studio Code

Visual Studio Code will be primarily used by the Information Technology Department for software development. It supports numerous programming languages, extensions, debugging tools, and development workflows.

The software is especially appropriate for the company's developers because it is lightweight and supports technologies commonly used for web and application development.

Git

Git provides version control for software development projects. It allows developers to track changes to source code, create branches, collaborate on projects, and restore previous versions when necessary.

Using version control reduces the risk of losing development work and improves collaboration among IT employees.

GitHub Desktop

GitHub Desktop provides a graphical interface for Git and GitHub workflows. It can make repository management easier for employees who are less familiar with command-line Git commands.

It can be used to clone repositories, commit changes, manage branches, and synchronize development projects with remote repositories.

VirtualBox

VirtualBox allows the IT Department to create and operate virtual machines on physical computers. This is useful for testing operating systems, applications, server configurations, and network environments without affecting production systems.

Virtualization also provides a safe environment for learning and testing system administration configurations.

Google Chrome

Google Chrome will be installed on employee computers as the company's primary web browser. Employees will require a browser to access websites, cloud services, web applications, email systems, online collaboration tools, and other internet resources.

The browser should be kept updated to reduce security vulnerabilities.

Microsoft Defender

Microsoft Defender provides built-in antivirus and endpoint protection for Windows computers. It helps detect and block malware, viruses, spyware, and other security threats.

The IT Department should ensure that Defender security intelligence and Windows security updates remain enabled and up to date.

AnyDesk

AnyDesk can be used by authorized IT personnel to provide remote technical support. It allows technicians to troubleshoot supported computers without always requiring physical access to the device.

Remote access should be strictly controlled and only enabled when necessary. Strong authentication and organizational policies should be used to prevent unauthorized remote access.

7-Zip

7-Zip is a file compression and archive management utility. Employees can use it to compress large files, extract downloaded archives, and package multiple files for transfer.

Because it is open-source and lightweight, it provides useful functionality without significant additional software licensing costs.

3.4 Software Deployment Strategy

Software will be deployed according to employee responsibilities and departmental requirements. Standard applications such as Windows 11 Pro, Microsoft 365, Google Chrome, Microsoft Defender, and 7-Zip can be installed on most employee computers.

Development and administration software such as Visual Studio Code, Git, GitHub Desktop, VirtualBox, and specialized system administration utilities will primarily be installed on computers assigned to the Information Technology Department.

The IT Department should maintain an updated software inventory and ensure that all applications receive security updates and patches. Unauthorized software installation should be restricted to reduce security risks and prevent software conflicts.

3.5 Software Management Recommendations

ABC Startup Solutions should establish centralized software management procedures to ensure that company computers remain secure and consistent.

The IT Department should:

- Maintain a software inventory for all company devices.
- Keep operating systems and applications updated.
- Remove unnecessary or unsupported software.
- Use properly licensed commercial software.
- Restrict unauthorized software installation.
- Regularly review software licenses and subscriptions.
- Maintain security software on all supported endpoints.
- Document important software configurations.
- Back up important application and configuration data.
- Periodically review software requirements as the company expands.

A properly managed software environment will help ABC Startup Solutions maintain secure, reliable, and productive IT operations while controlling unnecessary software costs.

PART 4 – ENTERPRISE NETWORK INVENTORY

4.1 Network Inventory Overview

ABC Startup Solutions requires a reliable and secure network infrastructure to connect its 20 employees, servers, printers, storage devices, and wireless devices. Since the company is starting its IT infrastructure from scratch, the network will be designed with security, performance, manageability, and future expansion in mind.

The proposed network will use a combination of wired Ethernet connections and secure wireless connectivity. Wired connections will be prioritized for desktop computers, servers, printers, and other critical equipment, while wireless access points will provide Wi-Fi connectivity for laptops and authorized mobile devices.

A firewall will be positioned between the internet connection and the internal network to provide an additional security layer. The managed switch will allow the IT Department to implement VLANs and logically separate departmental traffic.

4.2 Enterprise Network Inventory

Asset ID	Network Equipment	Quantity	Location / Department	Purpose
NET-001	ISP Modem / ONT	1	IT / Network Room	Receives the company's internet connection from the Internet Service Provider
NET-002	Business Router	1	IT / Network Room	Performs routing, DHCP, and communication between internal networks
NET-003	Network Firewall	1	IT / Network Room	Protects the internal network from unauthorized access and external threats
NET-004	24-Port Managed Gigabit Switch	1	IT / Network Room	Connects wired computers, server, printers, NAS, and wireless access points
NET-005	Wi-Fi 6 Wireless Access Point	2	Office Floor	Provides wireless network connectivity for laptops and authorized devices
NET-006	24-Port Patch Panel	1	IT / Network Room	Organizes and terminates network connections from office Ethernet outlets
NET-007	CAT6 Ethernet Cable	30+ runs	All Departments	Provides wired network connections between computers and network equipment
NET-008	RJ45 Connectors	50+	IT / Network Room	Used for terminating and repairing Ethernet cables
NET-009	Network Rack	1	IT / Network Room	Houses and organizes network equipment and cable connections

NET-010	UPS	2	IT / Network Room	Provides backup power and protection for critical network equipment

Part 5 – Enterprise Network Diagram

5.1 Network Topology Overview

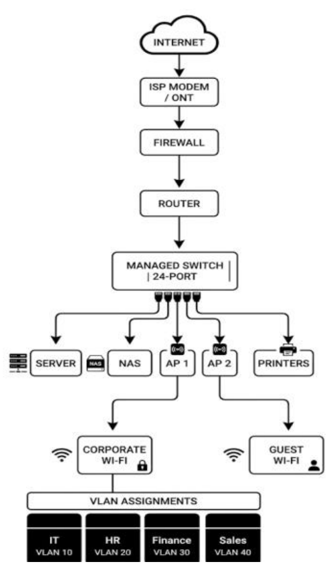
The proposed network topology for ABC Startup Solutions is designed to provide secure, reliable, and manageable connectivity for all departments. The network follows a hierarchical structure in which internet traffic passes through the ISP modem, firewall, router, and managed switch before reaching internal company resources.

The network uses a combination of wired and wireless connections. Desktop computers, servers, printers, and NAS storage are connected through the managed switch, while laptops and authorized wireless devices connect through the wireless access points.

Network segmentation is implemented using VLANs to separate departmental traffic and improve security.

5.2 Network Topology

The logical connection of the network is:



6.1 Helpdesk Technician

Responsibilities

A Helpdesk Technician provides first-level technical support to employees and helps resolve common hardware, software, and connectivity problems. In ABC Startup Solutions, the Helpdesk Technician would assist employees with computer issues, software installation, account problems, printer issues, network connectivity, and basic troubleshooting.

Typical responsibilities include:

- Responding to employee IT support requests
- Troubleshooting hardware and software problems
- Installing and configuring approved software
- Setting up employee computers and user accounts
- Troubleshooting printers and network connectivity
- Escalating complex problems to specialized IT personnel
- Maintaining records of reported problems and solutions
- Assisting with basic cybersecurity procedures

Skills

A Helpdesk Technician should have:

- Hardware and software troubleshooting skills
- Basic networking knowledge
- Windows administration skills
- Customer service and communication skills
- Problem-solving skills
- Basic cybersecurity awareness
- Documentation and ticket management skills

Common Tools

Common tools include:

- Windows 11 Pro
- Microsoft 365
- Google Chrome
- AnyDesk

- Microsoft Defender
- Remote support tools
- Helpdesk/ticketing systems
- Hardware diagnostic utilities

Certifications

Useful certifications include:

- CompTIA A+
- CompTIA Network+
- CompTIA ITF+
- Microsoft-related certifications

CompTIA A+ is particularly relevant because it covers fundamental hardware, operating system, troubleshooting, and IT support knowledge.

6.2 Network Administrator

Responsibilities

A Network Administrator is responsible for maintaining the organization's network infrastructure and ensuring that employees have reliable and secure network connectivity.

At ABC Startup Solutions, the Network Administrator would manage the firewall, router, managed switch, wireless access points, VLANs, and other networking equipment.

Typical responsibilities include:

- Configuring and maintaining network devices
- Managing IP addressing and DHCP
- Configuring VLANs
- Monitoring network performance
- Troubleshooting connectivity problems
- Managing wireless networks
- Maintaining network security configurations
- Performing network backups and documentation
- Monitoring network availability
- Planning network upgrades

Skills

A Network Administrator should have:

- TCP/IP networking knowledge
- Routing and switching skills
- VLAN configuration skills
- Firewall administration knowledge
- Wireless networking knowledge
- Network troubleshooting skills
- Network security awareness
- Network monitoring skills

Common Tools

Common tools include:

- Cisco or other managed switches
- Business routers
- Firewalls
- Wi-Fi access points
- Wireshark
- Ping
- Traceroute
- Network monitoring systems
- SSH
- Network configuration interfaces

Certifications

Useful certifications include:

- CompTIA Network+
- Cisco Certified Network Associate (CCNA)
- CompTIA Security+
- Vendor-specific networking certifications

The CCNA is particularly useful for professionals who want to develop stronger routing, switching, and networking skills.

6.3 Linux System Administrator

Responsibilities

A Linux System Administrator manages Linux-based servers and services. At ABC Startup Solutions, this role would primarily involve managing the company's Ubuntu Server environment.

The administrator would be responsible for maintaining server availability, configuring services, managing users, applying updates, monitoring system performance, and protecting server resources.

Typical responsibilities include:

- Installing and configuring Ubuntu Server
- Managing Linux user accounts and permissions
- Installing and updating packages
- Managing server services
- Monitoring CPU, memory, storage, and system performance
- Managing file systems and storage
- Configuring network services
- Performing server backups
- Troubleshooting server problems
- Applying security updates
- Reviewing system logs

Skills

A Linux System Administrator should have:

- Linux command-line knowledge
- Bash scripting skills
- File and permission management
- Networking knowledge
- Server security knowledge
- Storage management skills
- Troubleshooting skills

- System monitoring skills

Common Tools

Common tools include:

- Ubuntu Server
- SSH
- Bash
- systemd
- `top` / `htop`
- `df` and du
- `grep`
- `journalctl`
- `apt`
- Vim or Nano
- VirtualBox

Certifications

Useful certifications include:

- Linux Professional Institute (LPI) certifications
- CompTIA Linux+
- Red Hat Certified System Administrator (RHCSA)
- Linux Foundation certifications

These certifications can demonstrate practical knowledge of Linux administration and server management.

6.4 Cloud Administrator

Responsibilities

A Cloud Administrator manages cloud-based infrastructure, services, storage, users, and security configurations. As ABC Startup Solutions grows, cloud services may be introduced to supplement or replace some on-premises infrastructure.

Responsibilities may include:

- Managing cloud servers and virtual machines
- Configuring cloud storage
- Managing cloud user accounts and permissions
- Monitoring cloud resources
- Managing cloud backups
- Configuring virtual networks
- Implementing cloud security controls
- Monitoring cloud costs
- Supporting cloud-based applications
- Assisting with cloud migration projects

Skills

A Cloud Administrator should have:

- Cloud computing fundamentals
- Networking knowledge
- Virtualization knowledge
- Identity and access management skills
- Cloud security knowledge
- Linux and Windows administration skills
- Backup and disaster recovery knowledge
- Cloud monitoring skills

Common Tools

Common tools include:

- Microsoft Azure
- Amazon Web Services (AWS)
- Google Cloud
- Azure Portal
- AWS Management Console
- Cloud command-line interfaces
- Virtual machines
- Cloud monitoring tools
- Identity and access management systems

Certifications

Useful certifications include:

- Microsoft Certified: Azure Administrator Associate
- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect – Associate
- Google Associate Cloud Engineer

A cloud certification can help demonstrate knowledge of cloud infrastructure, security, networking, and resource management.

6.5 Collaboration Between IT Professionals

The Helpdesk Technician, Network Administrator, Linux System Administrator, and Cloud Administrator work together to maintain a reliable and secure IT environment. The Helpdesk Technician serves as the primary point of contact for employees and resolves common technical issues before escalating more complex problems. The Network Administrator maintains the company's network infrastructure, including the router, firewall, switch, VLANs, and wireless access points. The Linux System Administrator manages Linux servers and the services running on them, while the Cloud Administrator manages cloud-based resources and services as the company adopts cloud technologies. These professionals must communicate and coordinate with one another because many IT problems involve multiple areas of infrastructure. For example, an employee who cannot access an internal application may first contact the Helpdesk Technician, who can determine whether the problem is related to the employee's computer, network connection, server, or cloud service before escalating it to the appropriate specialist. Working together allows the organization to maintain availability, security, performance, and efficient IT operations.

6.6 Importance of Teamwork in System Administration

System administration is not limited to maintaining individual computers. Modern organizations depend on interconnected hardware, software, networks, servers, security systems, and cloud services.

For ABC Startup Solutions, cooperation between these IT professionals will help ensure that:

- Employees receive timely technical support.

- Network services remain available.
- Servers are properly maintained.
- Company data is protected.
- Security incidents are addressed quickly.
- Cloud resources are properly managed.
- IT problems are documented and resolved efficiently.
- The infrastructure can scale as the company grows.

Effective communication and clear documentation are therefore essential skills for every system administration professional.

PART 7 – INFRASTRUCTURE RECOMMENDATIONS

7.1 Overview

ABC Startup Solutions is a newly established software development and IT consulting company with 20 employees. Since the organization is building its IT infrastructure from the ground up, its initial infrastructure should prioritize **reliability, security, scalability, data protection, and cost efficiency**.

The following recommendations are based on the company's current size and expected future growth. The infrastructure should be capable of supporting current operations while allowing additional employees, devices, applications, and services to be added without requiring a complete redesign.

7.2 Internet Provider

ABC Startup Solutions should subscribe to a **business-grade fiber internet service** rather than relying on a basic residential internet plan.

A business connection should provide adequate bandwidth, better reliability, and technical support appropriate for an organization. For the company's initial 20 employees, a connection of approximately **300–500 Mbps** should be sufficient for normal activities such as web browsing, cloud applications, video conferencing, software development, file transfers, and remote support.

The company should also consider an internet service with:

- Business-grade technical support

- Static or dedicated IP availability
- Service-level agreements (SLA), when available
- Reliable fiber connectivity
- Sufficient upload bandwidth
- Options for future bandwidth upgrades

As the company grows, its internet plan should be reviewed based on actual network utilization.

7.3 Server Specifications

The company should deploy one dedicated business-class server running **Ubuntu Server LTS** for its initial infrastructure.

Recommended Server Specifications

Component	Recommended Specification
Processor	8-core server-grade CPU
Memory	32–64 GB ECC RAM
Storage	2–4 enterprise SSD/HDD drives
Storage Configuration	RAID 1 or RAID 5/6 depending on requirements
Network	Dual Gigabit Ethernet or faster
Power	Redundant power supply if budget permits
Operating System	Ubuntu Server LTS
Backup	Automated backup to NAS and external storage
Management	Remote server management capability

The server can initially host services such as internal applications, file sharing, development services, databases, and other company infrastructure services.

Server resources should be monitored regularly so that additional memory, storage, or processing capacity can be added before performance becomes a problem.

7.4 Backup Strategy

ABC Startup Solutions should implement a **3-2-1 backup strategy** to protect important business data.

The strategy should maintain:

- **3 copies** of important data

- **2 different types of storage media**
- **1 copy stored offline or off-site**

A possible implementation is:

Backup Location	**Purpose**
Production Server	Primary working data
NAS Storage	Local backup and recovery
External Backup Drive	Offline backup
Cloud / Off-site Storage	Protection against physical disasters

7.5 Security Recommendations

Security should be incorporated into the infrastructure from the beginning rather than added after deployment.

The company should implement the following controls:

Firewall

A dedicated firewall should protect the internal network from unauthorized external traffic.

VLAN Segmentation

The proposed VLAN structure should separate IT, HR, Finance, Sales, servers, and guest Wi-Fi.

This reduces unnecessary communication between departments and helps protect sensitive information.

Access Control

Employees should only receive access to the systems and information required for their jobs. Administrative privileges should be limited to authorized IT personnel.

Multi-Factor Authentication

MFA should be enabled for important accounts, particularly administrator accounts, cloud services, email, and other systems containing sensitive information.

Security Updates

Operating systems, applications, firmware, and network devices should be regularly updated to address known security vulnerabilities.

Physical Security

The server and network equipment should be placed in a restricted-access area. Only authorized personnel should have access to the network rack and server.

Security Monitoring

Network and server logs should be reviewed regularly to identify unusual activity, failed login attempts, and potential security incidents.

7.6 Antivirus and Endpoint Protection

All Windows computers should use **Microsoft Defender** or an appropriate business endpoint security solution.

Microsoft Defender provides built-in protection against malware and other common threats. The IT Department should ensure that real-time protection, security intelligence updates, and operating system updates remain enabled.

For stronger centralized management as the company grows, ABC Startup Solutions may eventually adopt a business endpoint detection and response (EDR) solution.

Employees should also be trained to recognize phishing emails, suspicious attachments, malicious links, and other common security threats because technical security controls alone cannot eliminate all risks.

7.7 Password Policy

ABC Startup Solutions should establish a formal password policy for all employees.

The policy should require:

- Long and unique passwords
- Different passwords for different systems
- Multi-factor authentication for important accounts
- No sharing of passwords
- No storing passwords in unsecured documents
- Immediate reporting of suspected account compromise
- Administrative accounts to use stronger authentication controls
- Use of an approved password manager where appropriate

Passwords should not be reused between personal and company accounts.

The company should also disable accounts promptly when employees leave the organization to prevent unauthorized access.

7.8 Physical Infrastructure and Power Protection

Critical networking and server equipment should be protected from power interruptions and electrical fluctuations.

The server, firewall, router, switch, and NAS should be connected to appropriately sized UPS units. The network room should also have adequate ventilation and should be kept free from unnecessary equipment and materials.

The network rack should use proper cable management and clearly labeled connections to simplify maintenance and troubleshooting.

7.9 Expansion Plan

The infrastructure should be designed to support future growth. ABC Startup Solutions currently has 20 employees, but the company may expand as its client base increases.

The following expansion strategy is recommended:

Phase 1 – Initial Deployment

Support the current 20 employees with:

- 14 desktop computers
- 6 laptops
- 1 server
- 1 NAS

- 1 managed switch
- 2 wireless access points
- VLAN segmentation
- Business-grade internet
- Dedicated firewall

Phase 2 – Moderate Growth

If the company reaches approximately 30–40 employees, it should consider:

- Additional managed switches
- Additional wireless access points
- Increased internet bandwidth
- Additional server resources
- Larger NAS capacity
- Additional network ports
- Centralized device management

Phase 3 – Larger Expansion

If the company grows significantly beyond its original office capacity, it should consider:

- Additional servers or virtualization infrastructure
- Cloud-based services
- Redundant network equipment
- Secondary internet connection
- More advanced security monitoring
- Additional office network segments
- Dedicated IT infrastructure staff

Planning for expansion now will reduce the cost and disruption associated with future infrastructure upgrades.

7.10 Overall Recommendation

ABC Startup Solutions should adopt a **secure, scalable, and manageable infrastructure** rather than focusing only on the lowest initial purchase cost.

The recommended combination of business-grade fiber internet, dedicated server infrastructure, NAS and offline backups, firewall protection, VLAN segmentation, endpoint security, strong authentication, UPS protection, and documented IT procedures will provide a solid foundation for the company's operations.

The infrastructure should also be reviewed regularly. As the organization grows, system administrators should monitor performance, security events, storage capacity, internet utilization, and employee requirements. This allows the company to upgrade its infrastructure based on actual business needs rather than replacing equipment prematurely.

PART 8 – PERSONAL REFLECTION

Personal Reflection

Working on the Enterprise Infrastructure Planning project helped me understand that system administration involves much more than troubleshooting computers and installing software. Before starting this project, I viewed IT infrastructure mainly as the hardware, software, and network devices used by an organization. Through this activity, I learned that a System Administrator must first understand the organization's requirements and then carefully plan the infrastructure before any equipment is purchased or configured.

One of the most important things I learned was how different components of an IT infrastructure work together. Creating the hardware and software inventories helped me understand how each device and application serves a specific purpose within the organization. I also learned the importance of network design, particularly the use of routers, firewalls, managed switches, wireless access points, and VLANs. Separating departments into different VLANs showed me how network segmentation can improve both security and network management.

The most challenging task for me was designing the enterprise network topology. It was necessary to determine how the Internet, ISP modem, firewall, router, switch, server, NAS, printers, wireless access points, and departmental computers should be connected. I also had to consider how the network could support the company's current requirements while allowing room for future expansion. This required me to think about the infrastructure as a complete system rather than as individual devices.

This project also taught me why planning is important before deployment. Purchasing equipment without first understanding the organization's requirements could result in unnecessary expenses, insufficient hardware, poor network performance, security problems, or difficulties when the company expands. Proper planning allows an organization to select appropriate hardware and software, establish security measures, create backup strategies, and design a network that can support business operations effectively.

I believe this project will help me become a better System Administrator because it improved my technical planning, documentation, problem-solving, and decision-making skills. It also showed me the importance of thinking about security, reliability, scalability, and business requirements when designing an IT infrastructure. In the future, I can apply these concepts when configuring real networks, managing servers, troubleshooting infrastructure, and planning IT solutions for organizations. Overall, this project gave me a better understanding of how professional system administrators plan and maintain the technology that businesses depend on every day.